

of Questions: 30

Question #: 1

A 54-year-old man comes to the physician because of a 1-week history of hearing loss in both ears. He has served in the military for the previous 30 years. Physical examination shows sensorineural hearing loss most likely caused by noise-induced trauma to the hair cells of the inner ear. Which of the following best characterizes the localization of these cells in the membranous labyrinth?

- A. Scala vestibuli
- ✓B. Scala media
- C. Scala tympani
- D. Utricle
- E. Sacculle

Rationale: The membranous labyrinth is divided into the vestibular labyrinth (which contains the utricle and saccule) and the cochlea labyrinth (which contains the cochlea duct). The vestibular labyrinth deals with balance while the vestibular labyrinth deals with sound production via the organ of Corti. Loss of the hair cells of the cochlea labyrinth will result in hearing loss. The hair cells are part of the organ of Corti, which is located on the floor (formed by the basilar membrane) of the scala media.

Item Description: HCB.1383 - Membranous labyrinth

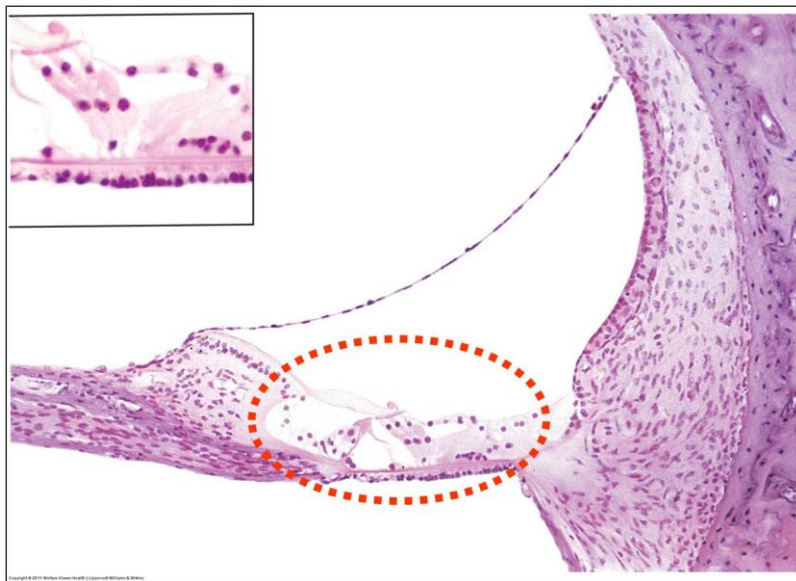
Question #: 2

An 8-year-old boy is brought to the physician by his parents because of a 3-month history of worsening hearing loss. Physical examination shows a sensorineural hearing loss of both ears. Urinalysis shows massive proteinuria. The physician suspects an autoimmune destruction to the circumscribed structure shown in the attached photomicrograph. Which of the following structures most likely forms part of the affected tissue?

- A. Spiral ganglion
- B. Vestibular membrane
- ✓C. Tectorial membrane
- D. Stria vascularis
- E. Otolithic membrane

Rationale: The circumscribed area is the organ of Corti. This organ is for hearing and consists of hair cells with their supporting phalangeal cells. These sensory hair cells sit on the basilar membrane. They are anchored in the tectorial membrane.

Attachment:



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Item Description: HCB.1383 - Structure of the Organ of Corti

Question #: 3

A 12-year-old boy is brought to the physician by his mother because of a 1-day history of pain in his throat when swallowing. Physical examination shows the posterior aspect of the tongue is swollen and red. Which of the following nerves is most likely responsible for the pain felt by the patient?

- A. Facial
- B. Lingual
- C. Internal laryngeal
- D. Vagus

✓E. Glossopharyngeal

Rationale: The posterior 1/3 of the tongue is innervated for both sensation and taste by the glossopharyngeal nerve.

Item Description: A 12-year-old boy is

Question #: 4

A 24-year-old woman comes to the physician because of a 3-week history of a continuous painful sensation. She says she has a full feeling in her throat while speaking. Physical examination shows no swelling or redness in the pharynx. Laryngoscopy shows a large growth on the left vestibular fold. The pain from the lesion is most likely transmitted by which of the following nerves?

- A. Recurrent laryngeal
- B. External laryngeal
- ✓C. Internal laryngeal
- D. Glossopharyngeal
- E. Pharyngeal plexus

Rationale: The left vestibular fold (false vocal cord) is located in the larynx above the true vocal cords. This area is innervated by the internal laryngeal nerve.

Item Description: A 24-year-old woman

Question #: 5

A 25-year-old man comes to the physician because of a 2-week history of decreased salivation and flattening of the nasolabial fold. Physical examination shows decreased taste on the anterior third of the tongue and decreased strength in his facial muscles. There are no problems with lacrimation and his nasal cavity is moist. There is no loss of hearing or problems with balance. At which of the following locations is the lesion that resulted in these symptoms most likely located?

- A. Facial nerve in the internal acoustic meatus
- ✓B. Facial nerve in the medial wall of the middle ear

- C. Facial nerve at the stylomastoid foramen
- D. Greater petrosal nerve in the pterygopalatine fossa
- E. Chorda tympani in the infratemporal fossa

Rationale: Based on the symptoms the nerve that was damaged is the facial nerve before the branching of the chorda tympani but after the greater petrosal nerve is given off.

Item Description: A 25-year-old man co

Question #: 6

An 18-year-old woman comes to the physician because of a 1-day history of being unable to speak. She attended a sports event 3 days ago during which she became hoarse. Physical examination shows no signs of fever or cough. The physician explains that her condition is due to strain on the muscle responsible for tensing the vocal cords. Which of the following nerves most likely innervates the affected muscle?

- A. Recurrent laryngeal
- ✓B. External laryngeal
- C. Internal laryngeal
- D. Superior laryngeal
- E. Glossopharyngeal

Rationale: The muscle responsible for tensing the vocal cords is the cricothyroid muscle which is innervated by the external laryngeal nerve.

Item Description: An 18-year-old woman

Question #: 7

A 2-month-old boy is brought to the emergency department by his parents because of a sudden onset of seizures. His mother says he has been feeding poorly for the past 2 days. His blood ammonia level is 800 µg/dL (normal range is 15 - 45 µg/dL). His blood urea nitrogen level is low, and urine and serum are positive for orotic acid. Which of the following enzyme deficiency disorders is most likely?

- A. Lysosomal glucosidase
- B. Glucose 6-phosphatase
- C. Medium chain acyl CoA dehydrogenase
- D. UMP synthase
- ✓E. Ornithine transcarbamylase

Rationale: The vignette describes a child with a urea cycle defect. Note the hyperammonemia. The presence of orotic acid in urine is diagnostic of hyperammonemia-type 2 characterized by deficiency of Ornithine transcarbamylase (OTC). It is an X-linked recessive disorder and is more common in males.

Item Description: BCHM.MB. 2001: Urea cycle disorder OTC

Question #: 8

A 10-month-old boy is brought to the emergency department by his parents because of a 2-hour history of seizures. Physical examination shows an enlarged liver and kidney. Examination of his heart is within normal limits. Laboratory studies are shown in the table. Urine samples are negative for reducing sugars. Which of the following inherited disorders of metabolism is the most likely diagnosis?

- A. Myopathic carnitine deficiency
- B. Phenylketonuria
- C. Classical galactosemia
- D. Hereditary fructose intolerance
- ✓E. von Gierke disease
- F. McArdle disease
- G. Medium chain acyl CoA dehydrogenase deficiency

Rationale: The vignette describes a patient with hypoglycemia with lactic acidosis. There is also hepatomegaly. The patient has von Gierke disease due to deficiency of liver glucose 6-phosphatase.

Attachment:

Analyte	Patient	Reference range
Glucose	45 mg/dL	70-100 mg/ dL
Lactate	4 mmol/ L	0.5-1 mmol/ L
Arterial pH	7.3	7.35-7.45
Ketone bodies	0.6 mmol/L	0.4-0.5 mmol/L

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Item Description: BCHM.MB.0910; von Gierke disease

Question #: 9

A 14-year-old girl is brought to the physician by her parents because of a 1-week history of increased frequency of urination, increased thirst and weight loss. Physical examination shows dry mucus membranes. Urine dipstick tests are positive for glucose and ketone bodies. Which of the following would most likely be an additional laboratory finding in this patient?

- A. Decreased plasma glucose levels
- B. Normal HbA1c levels
- C. Decreased glucose output by liver
- D. Increased circulating C-peptide levels
- ✓E. Increased anion gap metabolic acidosis
- F. Increased serum bicarbonate levels

Rationale: The vignette describes a patient with type 1 diabetic ketoacidosis which is characterized by very low or undetectable circulating insulin levels and low C-peptide levels.

The elevated levels of ketone bodies in the blood results in increased anion gap metabolic acidosis.

Ketone bodies are weak acids and lower serum bicarbonate levels.

Hepatic gluconeogenesis is responsible for the increased plasma glucose levels. Normally, gluconeogenesis in the liver is inhibited by insulin. HbA1c levels would most likely be elevated due to the hyperglycemia.

Item Description: Type -1 Diabetes mellitus

Question #: 10

A 45-year-old woman comes to the physician because of a 2-month history of reduced hearing in both ears. She was treated with a 2-week regimen of gentamicin (an aminoglycoside antibiotic known to cause vestibular or cochlear ototoxicity) 3 months ago due to a complicated urinary tract infection. Physical examination shows no abnormalities. Audiometry shows sensorineural hearing loss most likely caused by damage to the hair cells of the organ of Corti of the

inner ear. Which of the following best characterizes the affected structure?

- A. Lateral surface has both kinocilium and stereocilia
- B. Apical surface contains a single true kinocilium and stereocilia
- C. Apical surface lacks both kinocilium and stereocilia
- ✓D. Kinocilium is absent on the apical surface
- E. Possesses a kinocilium on the basal aspect

Rationale: Hair cells are epithelial mechanoreceptors of the vestibular and cochlear labyrinth. Stereocilia and a single kinocilium are found on the surface of these specialized cells, except in the cochlear labyrinth. The kinocilium determines depolarization or hyperpolarization of these sensory cells. The stereocilia is made up of actin filaments and the kinocilium is made up of the basal body.

Item Description: HCB.1384 - Structure of the Organ of Corti - sensory hair cells

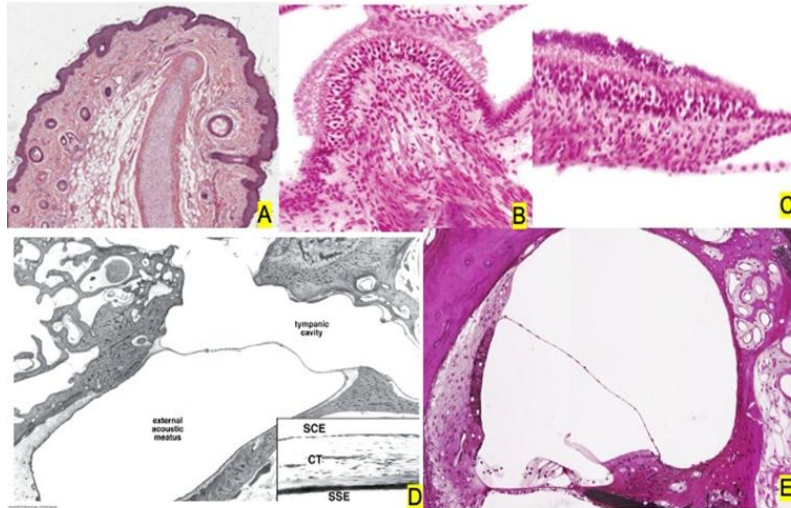
Question #: 11

A 5-year-old girl is brought to the physician by her mother because of a 1-week history of fever, cough, and discharge from the right ear. Her temperature is 40°C (104°F). Physical examination shows a febrile child with sunken eyes, poor skin turgor, and yellow, viscous discharge from the right ear. Otoscopy of the ear shows a perforation of the tympanic membrane, with a purulent discharge. Which of the following microanatomical photomicrographs most likely shows the perforated structure?

- A. A
- B. B
- C. C
- ✓D. D
- E. E

Rationale: The tympanic membrane separates the external acoustic meatus of the external ear from the tympanic cavity of the middle ear. This is best represented by option D in the images presented. Option A shows the pinna of the ear, option B shows the crista ampullaris of the semicircular ducts, option C shows the maculae of the utricle and saccule and option E shows the cochlea duct of the inner ear.

Attachment:



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Item Description: HCB.1360 - Tympanic membrane

Question #: 12

A 45-year-old man comes to the emergency department because of a 20-min history of bleeding in the scalp following a fall. His vital signs are within normal limits. Physical examination shows a 1-cm lacerated wound in the left temporal region of his scalp. The physician decides to suture this wound. He is administered lidocaine, a local anesthetic that blocks certain voltage-gated ion channels in the sensory nerves supplying the wound area. Which of the following ion fluxes during an action potential in the neuronal membrane is most likely blocked by this drug?

- A. K^+ influx
- B. K^+ efflux
- C. Na^+ efflux
- D. Cl^- efflux
- ✓ E. Na^+ influx

Rationale: Local anesthetics block voltage-gated sodium channels, thereby preventing action potentials being triggered and conducted. Hence the most significant impact is on sodium channels and the action potential. The resting membrane potential will be relatively unaffected.

Item Description: PHYS.0060 Action Potential block

Question #: 13

A 31-year-old woman comes to the physician because of an 8-month history of irregular menstrual cycles, acne, and darkening of the skin. Her blood pressure is 142/94 mm Hg; other vital signs are within normal limits. Physical examination shows increased hair on the face and hyperpigmentation of skin creases. Laboratory studies show elevated serum insulin, androgens and increased LH/FSH ratio. Pelvic ultrasonography shows cystic enlargement of the ovaries. She is diagnosed with polycystic ovarian syndrome. Which of the following is the most likely source of androgens during the early follicular stage in a normal woman?

- A. Cells of zona glomerulosa of adrenal cortex
- B. Granulosa cells in ovary
- ✓C. Theca cells in ovary
- D. Endometrial lining of the uterus
- E. Corpus luteum of the ovary

Rationale: The source of estrogen is granulosa cells. However, these cells get androgen precursors from the theca cells. Theca cells synthesize androgen from progesterone. This androgen moves into granulosa cells to synthesize estrogen. Therefore, theca cells are the source of androgens during early and mid cycle.

Item Description: PHYS.1076. Source of precursors for estrogen synthesis

Question #: 14

A 45-year-old man comes to the physician because of a 2-week history of headache, decreased urination, and muscle cramps. He was diagnosed with small cell carcinoma of the lung 4 months ago. His vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show elevated serum ADH and atrial natriuretic peptide. He is diagnosed with syndrome of inappropriate ADH (SIADH). Which of the following additional laboratory findings is most likely in this patient?

- A. Low plasma fasting glucose

- ✓B. Low serum osmolality
- C. Elevated serum sodium concentration
- D. Urinary osmolality less than 300 mOsm/kg
- E. Low serum potassium concentration

Rationale: In SIADH, there will be Hyponatremia and Euvolemia. Hyponatremia is because of retention of water without sodium. Low sodium reduces osmolality of serum because sodium is the major electrolyte that contributes to the total serum osmolality.

Euvolemia and normal blood pressure is due to the fact that the initial water retention results in volume expansion which activates secondary natriuretic mechanisms causing sodium and water loss in urine.

Item Description: PHYS.1021. SIADH lab findings

Question #: 15

A 34-year-old woman comes to the physician because of a 4-week history of abdominal discomfort and increased bowel movements. She says these symptoms are worse when she postpones her meals. Her vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies of a stool sample show an increased acidity. She is suspected of having deficiency of a hormone that normally helps to neutralize acid in the small intestine. A low level of which of the following hormones is most likely in this patient?

- A. Gastrin
- B. Histamine
- ✓C. Secretin
- D. Somatostatin
- E. Cholecystokinin

Rationale: The low stool pH points towards insufficient buffering of gastric acid in the small intestine. The key intestinal acidity buffering mechanism is bicarbonate release from pancreatic ductal cells, for which the main controlling hormone is secretin.

Item Description: PHYS.1130. Hormone involved in acid neutralization in small intestine

Question #: 16

A 45-year-old woman comes to the physician because of a 2-month history of difficulty hearing in the right ear. Neurological examination shows: Weber's test lateralizes to the right, Rinne's test BC is greater than AC on the right, AC is greater than BC on the left. Which of the following is the most likely location of the lesion?

- A. Inferior Colliculus
- B. Lateral Lemniscus
- C. Transverse temporal gyrus
- ✓D. Tympanic membrane
- E. Cochlear nerve

Rationale: The Weber's test and Rinne's test indicate that there is a lesion causing conductive hearing loss on the right side. Conductive hearing loss occurs when there is a lesion of the external ear or middle ear.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0244 Clinical assessment of auditory function

Question #: 17

A 62-year-old man comes to the physician because of a 2-day history of difficulty controlling the left side of his body. Imaging studies show an infarction affecting the left superior cerebellar peduncle. Which of the following motor systems is most likely dysregulated by the lesion?

- ✓A. Rubrospinal
- B. Lateral vestibulospinal
- C. Medial vestibulospinal
- D. Lateral reticulospinal
- E. Medial reticulospinal

Rationale: The red nucleus serves as a mesencephalic motor nucleus. It is regulated by the contralateral cerebellum via the superior cerebellar peduncle. The rubrospinal system is thus most likely affected by the lesion of the cerebello-rubral fibers. Note, however, that cerebello-thalamic fibers may also be affected.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0200 Cerebellar connections to

Question #: 18

A 27-year-old man comes to the emergency department because of a 2-day history of weakness in the lower extremities. He is recovering from a recent respiratory tract illness. Neurological examination shows absent deep tendon reflexes and symmetric weakness in both legs. Laboratory studies show elevated CSF protein and decreased nerve conduction velocity. Which of the following parts of the motor unit is most likely lesioned in this patient?

- A. Axon
- B. Extrafusal fibers
- C. Soma
- ✓D. Myelin sheath
- E. End plate

Rationale: Weakness in the lower limbs following a respiratory tract infection is commonly associated with Guillain Barre syndrome. This disease causes demyelination of the peripheral nerves, hence the hyporeflexia and decreased nerve conduction velocity. Guillain barre syndrome also commonly features elevated CSF protein.

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0184 Guillain-Barre

Question #: 19

A 62-year-old man comes to the physician because of a 2-hour history of traumatic injury to his left arm. Neurological examination shows manifestations consistent with lower motor neuron injury. Which of the following motor phenomena best represents a negative sign of the injury?

- A. Resting tremor
- B. Intention tremor
- ✓C. Hypotonia
- D. Fasciculations
- E. Waxing

Rationale: A negative sign is a sign in which a normally observable phenomenon diminishes (e.g., hypotonia reflects a loss of normal muscle tone). To the contrary, each of the other listed manifestations would constitute positive signs (e.g., fasciculations would reflect the emergence of twitching in persons who do not normally exhibit twitching).

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0180 Peripheral nervous disease manifestations

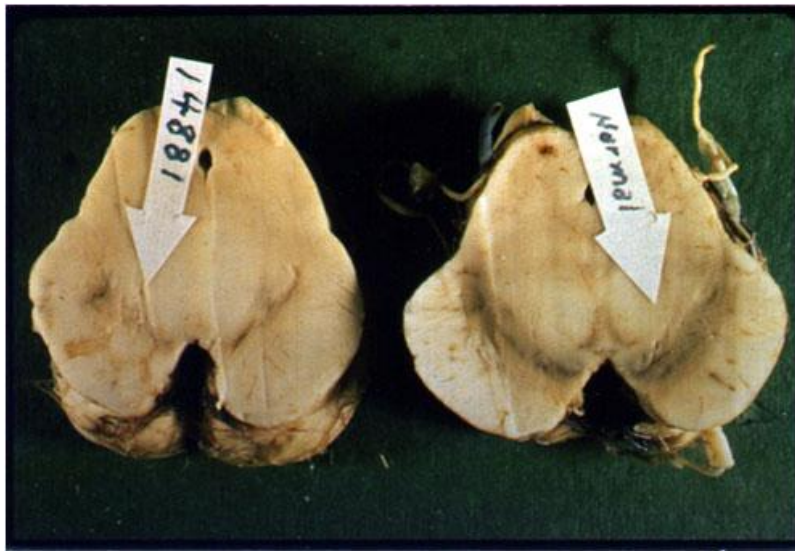
Question #: 20

A 58-year-old man comes to the physician because of an 8-month history of tremor. Neurological examination shows an expressionless face, resting tremor of the right upper extremity with rigidity. Movements of the right arm and hand are slow. The observed motor deficits most likely reflect degeneration of cells forming which of the following structures?

- A. Globus pallidus externus
- B. Globus pallidus internus
- C. Caudate nucleus
- D. Putamen
- ✓E. Substantia nigra pars compacta
- F. Substantia nigra pars reticulata
- G. Subthalamus

Rationale: Parkinson's disease is a hypokinetic disorder. It is due to insufficient release of dopamine from the Substantia nigra pars compacta. This results in decreased inhibition of the internal Globus pallidus and subsequent hypokinesia. Early in the disease the symptoms are observed on one side of the body. As the disease progresses, the symptoms become more debilitating and affect both sides of the body.

Attachment:



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Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0190 Parkinson direct and indirect

Question #: 21

A 53-year-old woman comes to the physician because of a 3-day history of weakness. Neurological examination shows decreased power and hyperreflexia in the right upper and lower limbs. Which of the following structures is most likely lesioned?

- A. Left internal capsule (anterior limb)
- B. Right internal capsule (genu)
- ✓C. Left crus cerebri
- D. Right crus cerebri
- E. Right posterior paracentral lobule
- F. Left anterior paracentral lobule

Rationale: The patient presents with upper motor neuron signs on the right side. The right side of the body is controlled by corticospinal fibers arising from the left side of the brain. These fibers arise in the left precentral gyrus and the left anterior paracentral gyrus. They travel through the posterior limb of the left internal capsule and left crus cerebri, decussate at the pyramids and enter the right side of the spinal cord where they communicate with lower motor neurons controlling the skeletal muscles on the right side of the body.

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0153 Correlate upper and lower

motor neuron signs with lesion sites

Question #: 22

A 34-year-old man comes to the physician because of a 4-day history of clumsiness affecting the right hand and foot. Physical examination shows right-sided dysdiadochokinesia and past-pointing. Which of the following additional manifestations is most likely?

- A. Athetosis
- B. Chorea
- C. Dystonia
- D. Rigidity
- ✓E. Intention tremor

Rationale: Athetosis, chorea, dystonia and rigidity are more commonly linked to basal gangliar disease. Cerebrocerebellar injuries commonly lead to intention tremor.

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 23

A 56-year-old man is brought to the physician by his spouse because of a 4-hour history of uncontrolled rotational movements of his left leg. A diagnosis of hemiballismus is made. Which of the following sites is most likely injured?

- A. Left frontal lobe
- B. Left striatum
- C. Left subthalamus
- D. Left cerebellum
- E. Left tectum
- F. Right frontal lobe
- G. Right striatum
- ✓H. Right subthalamus
- I. Right cerebellum

J. Right tectum

Rationale: Hemiballismus is a stroke-related, involuntary hyperkinesia characterized by sudden flinging, rotational movements of the arm and leg contralateral to a lesion in or near the subthalamic nucleus.

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0191 Hyperkinesia direct and indirect

Question #: 24

A 48-year-old man comes to the physician because of a 6-hour history of clumsiness. Neurologic examination shows dysarthria, right-sided dysdiadochokinesia, intention tremor, dysmetria and ataxia while attempting to perform skilled movements. Strength and sensory systems are normal. Which of the following cerebellar structures is most likely injured?

- A. Left vestibulocerebellum
- B. Left spinocerebellum
- C. Left cerebrocerebellum
- D. Right vestibulocerebellum
- E. Right spinocerebellum
- ✓F. Right cerebrocerebellum

Rationale: All of the clinical signs reflect cerebellar injury, most likely involving the lateral hemisphere. The cerebellum regulates movement of the ipsilateral body.

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 25

A 67-year-old man is brought to the physician because of a 16-year history of Parkinson disease. Which of the following manifestations is/are most likely in this patient?

- A. Dance-like movements
- ✓B. Bradykinesia

- C. Titubation
- D. Athetosis
- E. Intention tremor

Rationale: Parkinson's disease is a progressive neurodegenerative disease causing hypokinesia. Early in the disease the symptoms are observed on one side of the body. As the disease progresses, the symptoms become more debilitating and affect both sides of the body. Cardinal manifestations: resting tremor, bradykinesia, rigidity and postural instability.

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0190 Parkinson direct and indirect

Question #: 26

A 19-year-old woman comes to the physician because of a 3-week history of intermittent blurry vision. Neurologic examination shows an inability of her left eye to adduct during lateral gaze to the right. Both eyes converge normally. MRI of the brain shows an inflammatory lesion affecting the brainstem. Which of the following structures is most likely lesioned in this patient?

- A. Cranial nerve III
- B. Cranial nerve IV
- C. Cranial nerve VI
- ✓D. Medial longitudinal fasciculus
- E. Paramedian pontine reticular formation
- F. Pretectal area

Rationale: A lesion affecting the medial longitudinal fasciculus (MLF) will affect the adduction of the eye on the same side as the lesion during lateral gaze. Since both eyes converge normally, a cranial III lesion is less likely.

Item Description: SOM.MK.I.BPM2.1.NB.4.NSCI.0235 MLF lesion

Question #: 27

A 6-year-old boy is brought to the physician because of headaches and visual problems for 1 month. Imaging study shows an optic glioma compressing the right optic nerve. Which of the following is the most likely visual deficit caused by this

tumor?

- A. Hemianopia
- B. Homonymous anopia
- C. Heteronymous anopia
- D. Quadrantanopia
- ✓E. Monocular blindness

Rationale: Compression of the optic nerve would cause total loss of vision on the side of the lesioned nerve.

Item Description: SOM.MK.I.BPM2.1.NB.4.NSCI.0228 Visual Pathway Lesion

Question #: 28

A 40-year-old woman comes to the physician because of loss of smell that has lasted for 6 days. She has a history of Covid-19 infection and tested negative 4 days ago. Her vital signs are all within normal limits. Physical examination shows no abnormalities. Which of the following areas is most likely dysfunctional?

- A. Broca
- ✓B. Piriform
- C. Premotor
- D. Primary motor
- E. Somatosensory
- F. Wernicke

Rationale: The olfactory cortex is responsible for the affected sensation, and it includes the piriform cortex and entorhinal cortex.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0266 Cortical olfactory areas

Question #: 29

A 39-year-old man comes to the physician because of a 2-day history of blurry vision. Physical examination shows normal appearance of the right eye. Ptosis is present on the left. Neurological examination shows visual acuity of 20/20 in the right eye and 20/200 in the left eye (near card without glasses). Assessment of

eye movements shows injury to cranial nerve III. Imaging studies show a subarachnoid aneurysm that is compressing the cranial nerve as it exits the brainstem. Which of the following arteries is most likely expressing the aneurysm?

- A. Anterior cerebral
- B. Middle cerebral
- ✓C. Posterior communicating
- D. Anterior communicating
- E. Vertebral

Rationale: "Blurry vision" could mean many things, including double-vision from ocular misalignment, pupillary dilation and poor accommodation. Ptosis on the left can indicate a CN III lesion (eye lid opens weakly). Choice F approximates the bifurcation of the rostral basilar artery, which yields the posterior cerebral arteries at the junction of the pons and midbrain. The posterior communicating arteries arise nearby. This is the level of the ventral brainstem from which CN III emerges, rendering the nerve vulnerable to compression by an aneurysm blossoming from the posterior communicating artery.

Item Description: SOM.CS.I.BPM2.1.NB.4.NSCI.0234 Eye movement examination

Question #: 30

A newborn infant is assessed after birth for ambiguous, male-like genitalia. The infant becomes increasingly irritable over the next 4 hours. The blood pressure is remarkably low. Laboratory studies show hypoglycemia, hyponatremia and hyperkalemia. Karyotype studies show 46, XX. Which of the following enzymes is most likely deficient in this patient?

- A. 17-hydroxylase
- B. Aromatase
- ✓C. 21-hydroxylase
- D. 11-beta hydroxylase
- E. Renal 1-alpha hydroxylase

Rationale: The vignette describes a newborn infant with Congenital adrenal hyperplasia. The most common CAH is deficiency of 21-hydroxylase (CYP 21). This is characterized by decreased synthesis of mineralocorticoids and glucocorticoids.

Decreased mineralocorticoid synthesis results in hyponatremia (due to decreased Na⁺ reabsorption in the renal tubule) and low blood volume resulting in hypotension (low blood pressure).

Decreased glucocorticoid synthesis predisposes to hypoglycemia.

Increased diversion of steroid hormone precursors results in increased synthesis of adrenal androgens (Dehydroepiandrosterone sulfate; DHEAS) which result in enlargement of the clitoris in female infants and the appearance of ambiguous external genitalia.

Patients with 11-beta hydroxylase deficiency generally present with hypertension due to increased levels of 11-deoxycorticosterone which has mineralocorticoid activity. Increased sodium reabsorption may result in hypertension; Hypokalemia is usually an accompanying lab finding.

Patients with CYP 17 deficiency have hypertension. There is increased diversion of progesterone to form aldosterone. Females with the enzyme deficiency may have primary amenorrhea.

Item Description: Cong adrenal hyperplasia - CYP 21